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11.2.2 A strategy for computing the SVD

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applied it to the Spectral Decomposition problem, what we saw was that that algorithm is substantially more expensive than the QR factorization. Now we typically start with a matrix that has many more rows than it has columns. We can compute the QR factorization of that matrix. What we want you to do is to do a homework that links the Singular Value Decomposition of matrix A to the Singular Value Decomposition of

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Reading assignment

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Read Unit 11.2.2 of the notes. [\[LINK\]](#)

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Homework 11.2.2.1

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$, with $m \geq n$, and $A = QR$ be its QR factorization where, for simplicity, we assume that $n \times n$ upper triangular matrix R is nonsingular. If $R = \hat{U}\hat{\Sigma}\hat{V}^H$ is the SVD of R , give the Reduced SVD of A .

☒ done



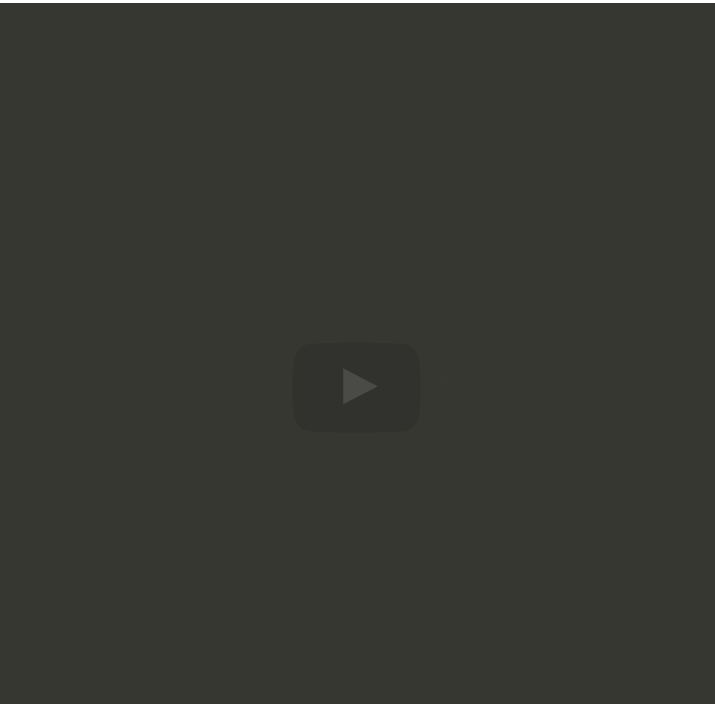
Solution

For a complete solution, see the [notes](#).

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Answers are displayed within the problem

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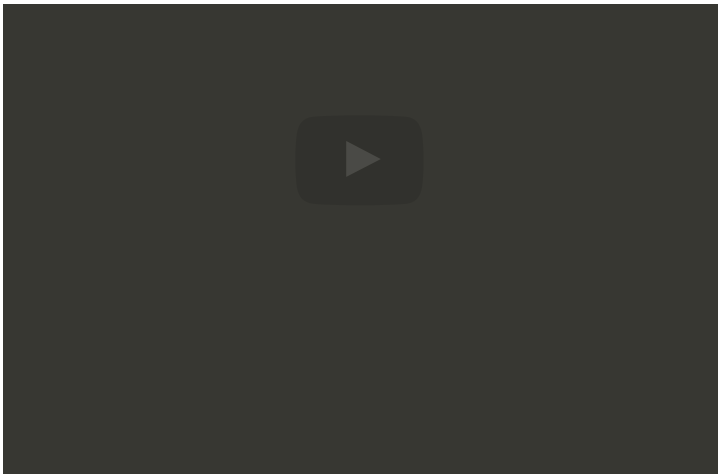
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Reduce Single Value Decomposition of
A. Well, more precisely, if there are some zeroes on the diagonal of Sigma _R, then we may have to get rid of a couple of columns in Q times U_R and in V_R to actually obtain the Reduce Singular Value Decomposition. But hopefully you get the point. So the important part is it pays the first compute the QR factorization of matrix A, especially if A has many, many more rows than it has columns. And then

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then combine that with our matrix Q. And with all that that we do it just right, we end up with something like the Reduced Singular



Value Decomposition of our original matrix A. Okay? Now, what's next? Well, we want to compute the Singular Value Decomposition now of matrix B. How might we do that? Well, we could form B Hermitian transpose times B.



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